Mauro Giliberti, Ph.D.

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Work Experience

2026 – current **PostDoc**, Goethe University Frankkurt.
CRC-TR211 collaboration, DFG funded.

Academic Background

2023 – 2025 **Ph.D. Università degli Studi di Firenze**, Physics and Astronomy department.
Thesis defended Mar 2026, Doctor Europaeus certification. Doctoral thesis: *Numerical Methods for High Energy Theoretical Physics and Astrophysics*.
Advisor: Aldo Lorenzo Cotrone.

2019 – 2022 **M.Sc. Università degli Studi di Firenze**, Scienze Fisiche e Astrofisiche - Fisica Teorica.
Graduated Dec 2022 with 110/110 cum laude.
Thesis: *Numerical Study of Bubble Dynamics in Holographic False Vacuum Decay*.
Advisor: Aldo Lorenzo Cotrone.
Co-advisors: Francesco Bigazzi, Bruno degli Esposti.

2016 – 2019 **B.Sc. Università degli Studi di Firenze**, Fisica e Astrofisica.
Graduated Oct 2019 with 107/110.
Thesis: *Effective One-Body Theory in General Relativity*.
Advisor: Domenico Seminara.

Publications

Articles

1. D. Chatzis, A. Fatemiabhari, M. Giliberti, and M. Hammond, "Holographic Entanglement Entropy in Quiver Theories," Sep. 2025. arXiv: 2509.19434 [hep-th].
2. Y. Bea, M. Giliberti, D. Mateos, M. Sanchez-Garitaonandia, A. Serantes, and M. Zilhão, "Bubble dynamics in a QCD-like phase diagram," Dec. 2024. arXiv: 2412.09588 [hep-th].
3. M. Giliberti, A. Fatemiabhari, and C. Nunez, "Confinement and screening via holographic Wilson loops," *JHEP*, vol. 11, p. 068, 2024. DOI: 10.1007/JHEP11(2024)068. arXiv: 2409.04539 [hep-th].

Software

1. M. Giliberti and B. Degli Esposti, *Numerical optimization with splines in high energy physics*, 2026. arXiv: 26XX.XXXXX.
2. M. Giliberti, *Robinhood.jl*, 2024. DOI: 10.5281/zenodo.13758659.

Fellowships and Scholarships

2025 – 2026 Università degli Studi di Firenze - "Marco Ademollo fellowship for Theoretical Physics" - Project PAGVARIDIPA.

2023 – 2025 European Union - Next Generation EU - National Recovery and Resilience Plan (NRRP) - M4C2 CN1 Spoke2 - Research Programme CN00000013 "National Centre for HPC, Big Data and Quantum Computing" - CUP B83C22002830001.

Experience

Talks, Posters and Scientific Visits

2025
- **Seminar** BCTP Bonn.
- **Visit and seminar** ITP Frankfurt.
- **Visit and seminar** ITP Utrecht.
- **Visit and seminar** HIP Helsinki.
- **Poster** "Simulating bubbles in a Holographic QCD-like phase diagram", ICHA4, Baku (Best Poster award).
- **Talk** "Numerical solution of holographic observables in quiver theories", AANL Yerevan.
- **Poster** "Simulating bubbles in a Holographic QCD-like phase diagram", Supercalcolo, Data Science e AI: dall'innovazione al trasferimento tecnologico, Firenze (Best Poster award).
- **Talk** "Numerical Methods for Holography", New Frontiers in Theoretical Physics - XXXVIII, Cortona.

2024
- **Talk** "Supercomputing and Big Data: new technology for new research horizons (Fundamental Research and Space Economy)", Festival della Scienza di Genova.
- **Poster** "Simulating the HQCD phase diagram", COST Action 22113 kick-off meeting, Padova.
- **Talk** "Why do we need a supercomputer to study boiling water?", Workshop «L'Università di Firenze al passo con il futuro del supercalcolo e dei big data», Firenze.

2023
- **Poster** "Holographic vacuum decay: a numerical study", HPC Summer School, Pavia.
- **Visit** ICCUB Barcelona.
- **Fire Talk** New Frontiers in Theoretical Physics - XXXVII, Cortona.

Conference and School Attendance

2026
- **CRC Retreat**, GSI Bonn.

2025
- **Julia and the Jecco module**, ICCUB.
- **Second Holography and Dense Matter Workshop**, APC Paris.
- **International Conference on Holography and its Applications**, Damghan University.
- **New Frontiers in Theoretical Physics - XXXVIII**, INFN.
- **AI4Phys workshop on "Artificial Intelligence for Physical Sciences"**, GGI.

2024
- **COST Action 22113 kick-off meeting**, UniPd.
- **Julia High Performance**, CINECA.

2023
- **XIX AVOGADRO MEETING on Strings, Supergravity and Gauge Theories**, UniPd.
- **LACES Winter School**, GGI.
- **Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles**, ICCUB.
- **New Frontiers in Theoretical Physics - XXXVII**, INFN.
- **HPC Summer School**, UniPv and CINI consortium.
- **Lectures on the Theory of Fundamental Interactions**, GGI.

2022
- **Reconstructing the Gravitational Hologram with Quantum Information**, GGI.

Teaching and Outreach

2025
- **Participation at FameLab selections**, Ferrara.
  Presenting research topics to the general public, winning the audience award.
- **Assistant professor for Physics**, UniFi.
  Tutoring physics, optics, and computer science bachelor students about mechanics, electrodynamics, and quantum physics.

Experience (continued)

2024 **Moderator for the *Strings in Florence* Journal Club**, UniFi.
Moderating and scheduling the weekly Journal Club, arranging external speakers, as well as discussion and dissertation activities of the team.

2023 **Assistant professor for Physics**, UniFi.
Tutoring engineering bachelor students about mechanics and electrodynamics.

2022 **Substitute teacher for maths and physics**, Liceo Scienfifico Leonardo da Vinci.
Teaching high-school level maths and physic.

2021 **Computer tutor**, UniFi.
Helping students and professors with the newly installed computers in the University's classrooms, about AV, networking, and conference call issues.

2020 **Guide at Museum of Mathematics**, Il Giardino di Archimede.
Showing the different exhibitions of physics and mathematics held at the museum to school groups, Italian and international, of age 10-18.

Skills

Languages	Italian native speaker, fluent in English, leaning Spanish and Japanese
Coding	MATLAB, Julia, Mathematica, C, C++, Python, LaTeX, Git
HPC	Slurm Workload, Leonardo by CINECA, Machine Learning for Scientific Applications
Misc.	Scouts leader in the AGESCI scouting movement (volunteer)

References

Aldo Lorenco Cotrone
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David Mateos
Professor
Universitat de Barcelona
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Carlos Núñez
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Swansea University
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